

Fold-dependent slowing in mechanistic CSD

This supplement collects manuscript-adjacent details that are useful for review but too long for the main text: the full paired geometry sweep behind the mechanistic slowdown statistics, the exact representative-run outputs underlying manuscript Table 1, the atlas-patch sanity-check figure, and a narrow physiology-anchor check against published CSD speed and delay bands. The physiology-anchor items come from the reduced biophysical validation workflow and should be read as contextual support rather than as a direct fit of the multi-ion mechanistic surface model.

Supplementary Tables

The speed bands summarized in Table S3 and Fig. S1 are the same published experimental and clinical ranges discussed in the main manuscript. In the present supplement they are included only as a narrow contextual check that accepted scalar and tensor validation cases remain within published speed regimes while preserving the tensor-faster-than-scalar ordering.

Table S1: Full paired geometry sweep for the mechanistic surface electrodiffusion model. Each row compares the same geometry with dipole alignment disabled or enabled on the standardized sweep mesh. The flat control is included as an internal specificity check and shows zero slowing. The depth 2.4 mm, $\sigma = 1.5$ mm sweep row is not the exact higher-resolution representative run, which is reported separately in Table S2.

Depth (mm)	σ (mm)	Severity	Vertices	Faces	Baseline speed (mm/min)	Dipole speed (mm/min)	Slowdown (mm/min)	Baseline delay (s)	Dipole delay (s)	Delay increase (s)	Baseline $ V_e $ (mV)	Dipole $ V_e $ (mV)
0.0	1.5	0.000	1248	2346	3.375	3.375	0.000	69.0	69.0	0.0	15.11	15.11
1.2	1.1	1.091	1248	2346	2.396	2.273	0.123	132.7	139.9	7.2	17.22	17.69
1.2	1.5	0.800	1248	2346	2.421	2.276	0.145	127.6	136.9	9.3	17.06	17.69
1.2	1.9	0.632	1248	2346	2.476	2.410	0.066	121.9	125.2	3.3	16.87	17.37
1.8	1.1	1.636	1248	2346	2.653	2.512	0.141	132.7	140.4	7.7	17.36	17.83
1.8	1.5	1.200	1248	2346	2.610	2.456	0.154	127.4	136.1	8.7	17.16	17.80
1.8	1.9	0.947	1248	2346	2.626	2.524	0.101	121.6	126.4	4.8	16.92	17.02
2.4	1.1	2.182	1248	2346	2.979	2.816	0.163	132.7	140.6	7.9	17.51	17.89
2.4	1.5	1.600	1248	2346	2.865	2.703	0.162	127.1	135.1	8.0	17.25	17.91
2.4	1.9	1.263	1248	2346	2.806	2.741	0.065	121.1	123.9	2.8	16.99	17.54
3.0	1.1	2.727	1248	2346	3.352	3.174	0.178	132.6	140.6	8.0	17.64	17.89
3.0	1.5	2.000	1248	2346	3.140	2.872	0.268	126.7	138.2	11.4	17.35	17.58
3.0	1.9	1.579	1248	2346	2.971	2.900	0.070	120.6	123.2	2.6	17.08	17.65

Table S2: Exact representative-run outputs underlying manuscript Table 1 for the folded surface used in the main manuscript (fold depth 2.4 mm, $\sigma = 1.5$ mm). Values are copied directly from the saved final representative run, not from a rerun. The full coupled run is retained only as an exploratory sensitivity analysis because vascular feedback changes the excitability regime.

Case	dt (s)	Speed (mm/min)	E1 arrival (s)	E2 arrival (s)	Delay (s)	Max $ V_e $ (mV)	Max K_e (mM)	Max swelling	Min oxygen	Min perfusion
Baseline mechanistic	0.04996995735874618	2.6091699831548576	31.431103178651348	159.40416397440032	127.97306079574898	15.433485430844861	31.521411772375073	0.8949639642825417	1.0	0.6
Dipole-aligned mechanistic	0.04996995735874618	2.5368718587402426	31.431103178651348	162.8520910321538	131.42098785350245	16.384820032184784	31.504237947379163	0.8949607589870553	1.0	0.6
Full coupled sensitivity	0.04996995735874618	3.5922541853540277	31.181253391857616	124.07540412176677	92.89415072990914	11.44262673137779	31.440285051723173	0.8948427323094323	0.2	0.25

Table S3: Accepted-candidate physiology anchor from the reduced biophysical validation sweep. This table is included as a contextual range check against published CSD speed and delay bands, not as a direct validation of the multi-ion mechanistic surface model.

Metric	Range (mm/min)	Mean	Within 2.0–5.0	Within 1.7–9.2	Within 2.5–4.5
Control	3.000–3.000	3.000	63/63	63/63	63/63
Scalar	2.606–2.750	2.635	63/63	63/63	63/63
Tensor	2.683–2.792	2.706	63/63	63/63	63/63
Tensor - scalar	0.042–0.077	0.071	–	–	–
Scalar downstream delay (s)	0.0–16.2 s	12.8 s	Within 10–30 s band: 54/63		

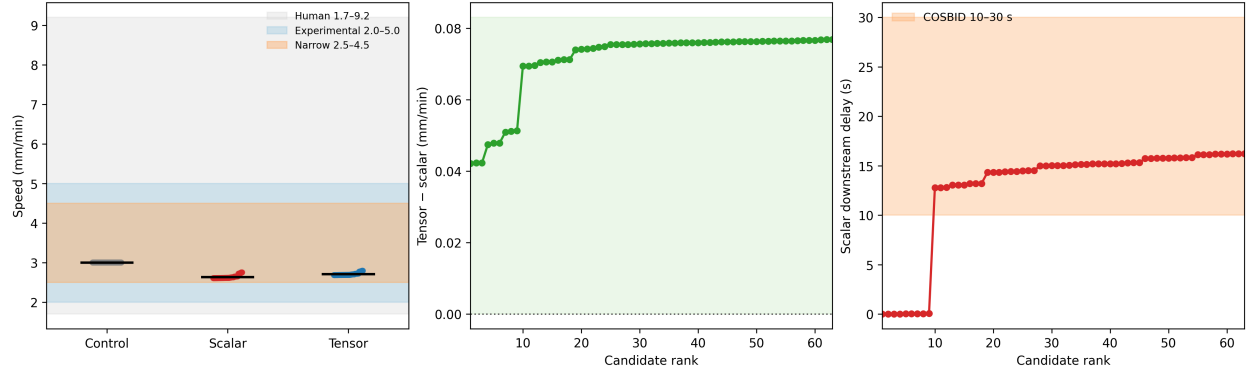


Figure S1: Physiology-anchor summary for the accepted reduced-biophysical validation cases. Shaded or annotated ranges denote the same broad published CSD speed and delay bands discussed in the main manuscript. The figure is included as a contextual supplement and should not be interpreted as a direct waveform-level validation of the mechanistic surface electrodiffusion model.

Supplementary Figures

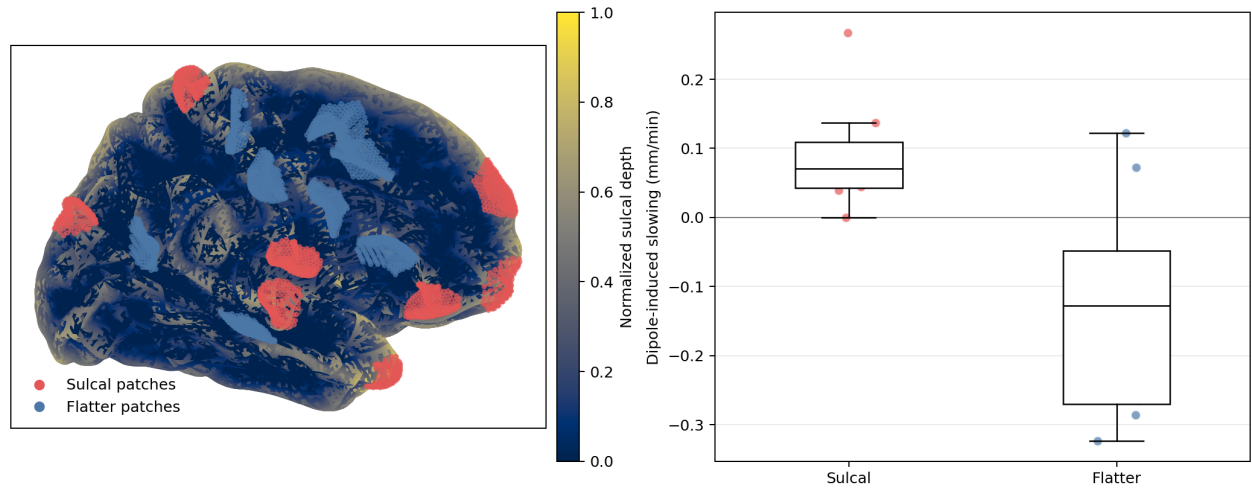


Figure S2: Atlas-based multi-patch sanity check used as a geometry-specific supplement to the main mechanistic results. Left, the fsaverage6 left-hemisphere midthickness surface with the $N = 8$ selected sulcal and $N = 8$ flatter matched patches overlaid on the normalized sulcal-depth map. Right, the distributions of dipole-induced propagation slowing for each patch category. The consistent positive slowdown on sulcal patches supports the main mechanistic results, whereas flatter patches (which still possess local structural curvature) yield inverted effects. This figure remains supplementary because it functions as an atlas-based sign-check rather than a full subject-specific human validation study.